

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1016	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	Joanathan Packia Singh	Reg. No.:	192472229
Date:	24/7/26	Duration:	60 minutes

Code of Conduct

I, Joanathan Packia Singh 192472229 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
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Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

Agile Sprint Planning & User Story Workshop

Fitness and Diet Tracker with Personalized Recommendations

Course Code:	CSA1016	Course Title:	Software Engineering
Student Name:	Joanathan Packia Singh 192472229	Reg. No.:	<u>192472229</u>

Code of Conduct: I certify that this submission is my original work and adheres to the academic integrity guidelines specified for this assessment. Signature: Joanathan

Agile Sprint Planning Approach

This document applies the Scrum framework to the Fitness and Diet Tracker problem statement. The team moves through backlog refinement, sprint planning, execution, review, and retrospective in a repeating cycle, with this workshop representing the Sprint Planning stage for Sprint 1. The sections that follow work through each planning artifact in sequence: project vision, epics and stories, acceptance criteria, backlog prioritization, sprint backlog, story point estimation, and the sprint goal.

Figure 1 — Scrum Sprint Cycle Applied to This Workshop

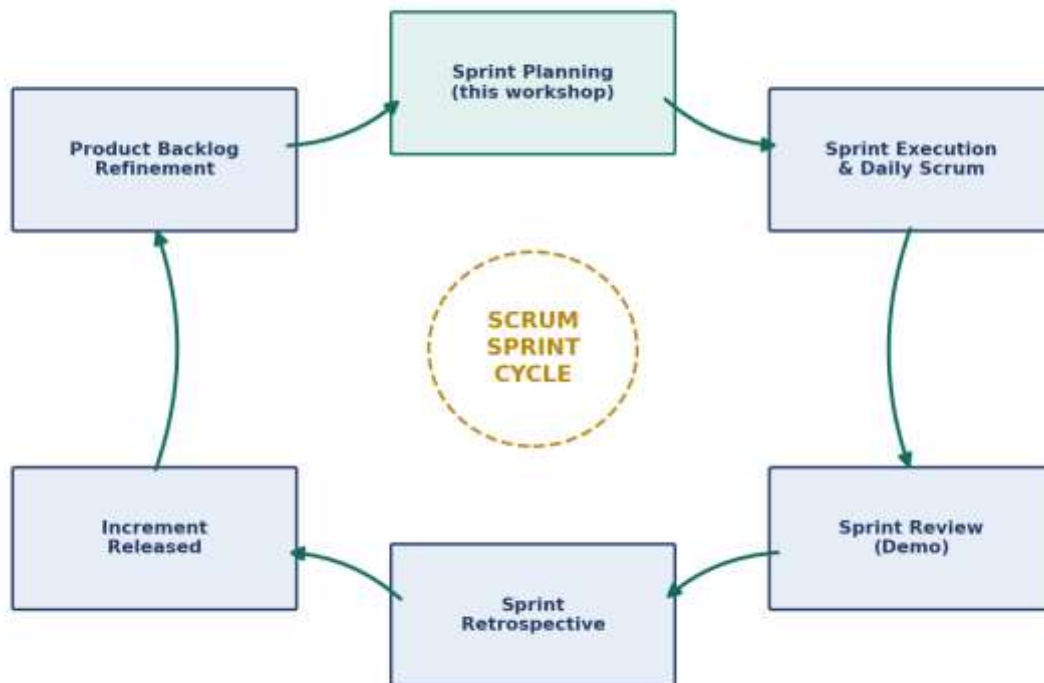


Figure 1: Scrum sprint cycle applied to this workshop — Sprint Planning is highlighted as the stage this document covers.

1. Project Vision and Stakeholders

1.1 Project Vision Statement

For health-conscious individuals who struggle to maintain consistent fitness and dietary habits, the Fitness and Diet Tracker is a mobile and web-based wellness platform that delivers personalized, AI-driven workout and nutrition recommendations based on each user's age, weight, BMI, and goals. Unlike existing fitness apps that offer generic plans and shallow progress tracking, our product combines wearable device synchronization, interactive dashboards, and continuous behavioral reminders to close the personalization gap and drive long-term adherence to healthy lifestyles.

1.2 Project Objectives

- Develop a personalized fitness and diet management platform accessible via mobile and web.
- Enable users to log and track daily physical activities and nutritional intake accurately.
- Generate AI-based, data-driven workout and diet recommendations tailored to individual health profiles.
- Integrate with wearable devices for automatic synchronization of health and activity metrics.
- Visualize user progress through interactive dashboards and periodic reports.
- Send timely reminders for workouts, hydration, and meals to improve engagement.
- Ensure secure storage and controlled sharing of personal health information.
- Promote sustained healthy lifestyle habits through continuous monitoring and feedback loops.

1.3 Key Stakeholders

Stakeholder	Description	Role / Interest in Project
End Users (Fitness & Diet Consumers)	Individuals tracking activity, calorie intake, and health goals	Daily app usage, data input, following recommendations, providing feedback
Nutritionists / Dietitians	Professionals validating diet plans and nutritional guidance	Review AI-generated diet plans, provide professional oversight, flag unsafe recommendations
Fitness Trainers / Coaches	Professionals guiding workout plans	Validate workout recommendations, adjust plans for injury/health conditions
Healthcare Providers	Doctors/clinics using health data for preventive care	Access authorized health summaries, monitor high-risk users, integrate clinical advice
Product Owner	Represents business and user interests	Owns the product backlog, prioritizes features, accepts/rejects completed work
Scrum Master	Facilitates the Agile process	Removes impediments, facilitates ceremonies, coaches the team on Agile practices
Development Team	Engineers, designers, QA, and data scientists building the product	Design, develop, test, and deliver increments; estimate story points
Wearable Device Partners	Third-party vendors (Fitbit, Garmin, Apple Health, Google Fit)	Provide APIs/SDKs for automatic data synchronization
System Administrators	Manage infrastructure, security, and data compliance	Ensure uptime, enforce data-privacy compliance and backup policies
Business & Marketing Team	Drives adoption, monetization, and partnerships	Define subscription tiers, promotions, and gym/nutritionist partnerships

2. Epics and User Stories

Epic 1 — User Onboarding & Profile Management

Enable new users to register, build a health profile, and configure personal fitness goals.

US-01 As a new user, I want to register using my email or social account, so that I can quickly create my personal account.

US-02 As a user, I want to input my age, weight, height, and BMI, so that the system can calculate my baseline health metrics.

US-03 As a user, I want to select my fitness goal (weight loss, muscle gain, maintenance), so that recommendations align with my objective.

US-04 As a user, I want to specify dietary preferences and restrictions (vegan, allergies, medical conditions), so that diet plans are safe and relevant to me.

Epic 2 — Activity & Nutrition Tracking

Allow users to log daily exercise, meals, and hydration to build an accurate activity history.

US-05 As a user, I want to log my daily workouts (type, duration, intensity), so that my activity history is up to date.

US-06 As a user, I want to log meals and snacks with calorie and macronutrient breakdown, so that I can monitor my nutritional intake.

US-07 As a user, I want to log my daily water intake, so that I stay aware of my hydration levels.

US-08 As a user, I want to search a food database when logging meals, so that I can quickly find accurate nutritional information.

Epic 3 — Personalized AI Recommendations

Use AI/ML analytics on health data to generate individualized workout and diet plans.

US-09 As a user, I want to receive a personalized weekly workout schedule, so that my exercise routine matches my fitness level and goal.

US-10 As a user, I want to receive a personalized daily meal plan, so that my nutrition supports my fitness goal.

US-11 As a user, I want the system to adjust recommendations automatically as my metrics change, so that my plan stays relevant over time.

US-12 As a nutritionist, I want to review and approve AI-generated diet plans flagged as high-risk, so that unsafe recommendations are prevented.

Epic 4 — Wearable Device Integration

Synchronize step count, heart rate, sleep, and calories burned from third-party wearables.

US-13 As a user, I want to connect my wearable device (Fitbit/Garmin/Apple Health), so that my activity data syncs automatically.

US-14 As a user, I want my heart rate and sleep data to sync in real time, so that recommendations account for my recovery and rest.

US-15 As a user, I want to be notified if my wearable device disconnects or fails to sync, so that I can reconnect it promptly.

Epic 5 — Progress Visualization & Dashboards

Provide users with visual insight into trends, milestones, and goal adherence.

US-16 As a user, I want to view an interactive dashboard summarizing my weekly activity and nutrition, so that I can track my progress at a glance.

US-17 As a user, I want to see trend graphs of weight, calories, and workouts over time, so that I can evaluate long-term progress.

US-18 As a user, I want to export a progress report as PDF, so that I can share it with my trainer or doctor.

Epic 6 — Reminders & Notifications

Increase adherence through timely, configurable behavioral nudges.

US-19 As a user, I want to receive reminders for scheduled workouts, so that I don't miss planned exercise sessions.

US-20 As a user, I want to receive hydration and meal reminders, so that I maintain consistent daily habits.

US-21 As a user, I want to customize reminder frequency and quiet hours, so that notifications don't disrupt my routine.

Epic 7 — Data Privacy & Security

Protect sensitive health data and give users control over how it is stored and shared.

US-22 As a user, I want my health data encrypted at rest and in transit, so that my personal information stays protected.

US-23 As a user, I want to control which third parties (nutritionist, doctor) can view my data, so that I retain ownership of my information.

US-24 As a system administrator, I want audit logs of data access, so that I can detect and respond to unauthorized activity.

Figure 2 — Backlog Hierarchy: Epics Decomposed into User Stories

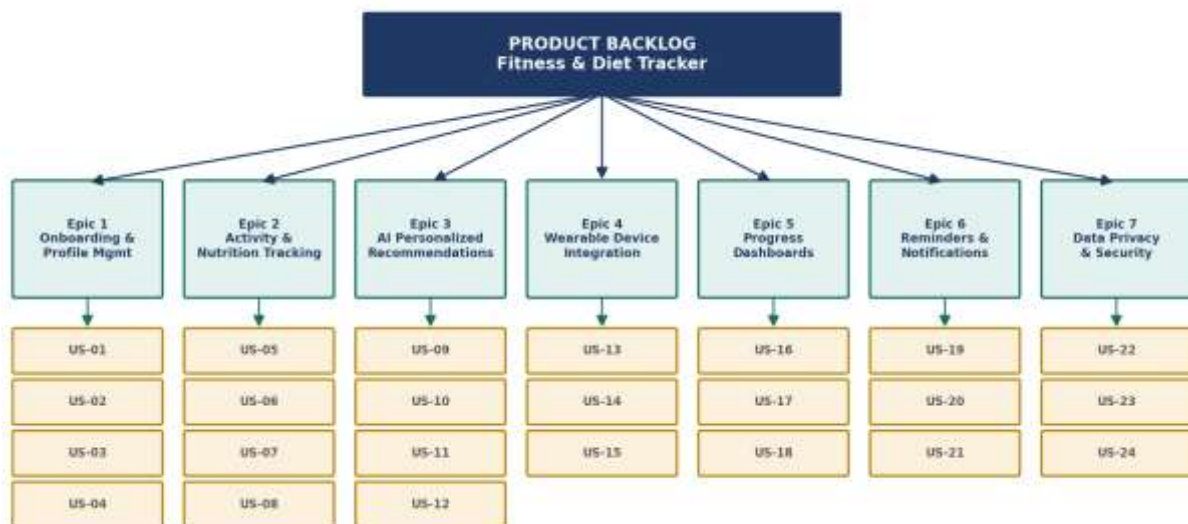


Figure 2: Backlog hierarchy showing all 7 epics decomposed into their 24 user stories.

3. Acceptance Criteria

Acceptance criteria for the highest-priority user stories selected for Sprint 1 are defined below using the Given-When-Then format to ensure each story is specific, measurable, and testable.

US-02: *As a user, I want to input my age, weight, height, and BMI, so that the system can calculate my baseline health metrics.*

Scenario 1 — Given a registered user is on the profile setup screen, **When** they enter valid age, weight, and height values, **Then** the system calculates and displays BMI along with a health category (underweight/normal/overweight/obese).

Scenario 2 — Given a user enters a non-numeric or out-of-range value, **When** they attempt to submit the form, **Then** the system displays a validation error and blocks submission until corrected.

US-05: *As a user, I want to log my daily workouts, so that my activity history is up to date.*

Scenario 1 — Given a user is on the activity log screen, **When** they select a workout type, duration, and intensity and save, **Then** the entry appears in today's activity log and updates the daily calories-burned total.

US-06: *As a user, I want to log meals with calorie and macronutrient breakdown, so that I can monitor my nutritional intake.*

Scenario 1 — Given a user searches the food database for an item, **When** they select a food and specify a portion size, **Then** the system adds the calories, protein, carbs, and fat to the user's daily nutrition total.

US-09: *As a user, I want to receive a personalized weekly workout schedule, so that my exercise routine matches my fitness level and goal.*

Scenario 1 — Given a user has completed onboarding with a stated fitness goal, **When** the recommendation engine runs its weekly scheduling job, **Then** a 7-day workout plan matching the user's goal, fitness level, and available equipment is generated and displayed.

Scenario 2 — Given a user's logged performance falls behind the current plan for 3+ consecutive days, **When** the weekly recalculation runs, **Then** the system adjusts intensity or reschedules missed sessions automatically.

US-10: *As a user, I want to receive a personalized daily meal plan, so that my nutrition supports my fitness goal.*

Scenario 1 — Given a user has set dietary preferences and a calorie target, **When** the diet recommendation engine generates a plan, **Then** the meal plan excludes restricted ingredients and stays within $\pm 5\%$ of the user's daily calorie target.

US-13: *As a user, I want to connect my wearable device, so that my activity data syncs automatically.*

Scenario 1 — Given a user selects "Connect Device" and authenticates with a supported wearable provider, **When** authorization succeeds, **Then** the app pulls the last 24 hours of steps, heart rate, and sleep data and displays a "Connected" status.

Scenario 2 — Given a connected device fails to sync for more than 6 hours, **When** the background sync job checks device status, **Then** the user receives a push notification prompting reconnection.

US-16: *As a user, I want to view an interactive dashboard summarizing my weekly activity and nutrition, so that I can track my progress at a glance.*

Scenario 1 — Given a user with at least one week of logged data opens the dashboard, **When** the dashboard loads, **Then** it displays total workouts, average daily calories, hydration percentage, and goal-progress indicators for the past 7 days.

US-19: *As a user, I want to receive reminders for scheduled workouts, so that I don't miss planned exercise sessions.*

Scenario 1 — Given a user has a workout scheduled for a specific time, **When** the scheduled time is 30 minutes away, **Then** the app sends a push notification reminding the user of the upcoming workout.

US-22: *As a user, I want my health data encrypted at rest and in transit, so that my personal information stays protected.*

Scenario 1 — Given a user submits health data through the app, **When** the data is transmitted to the server and stored in the database, **Then** it is encrypted using TLS 1.2+ in transit and AES-256 at rest, verified via a security audit log entry.

4. Product Backlog Prioritization

The product backlog below is prioritized using the MoSCoW technique (Must / Should / Could / Won't Have) combined with a Business Value score (1–10, based on stakeholder impact and alignment to project objectives). Stories are ranked to feed directly into sprint planning.

Story ID	User Story (Summary)	MoSCoW Priority	Business Value	Backlog Rank
US-01	Register using email or social account	Must Have	9	1
US-02	Input age, weight, height, BMI	Must Have	9	2
US-03	Select fitness goal	Must Have	8	3
US-05	Log daily workouts	Must Have	9	4
US-06	Log meals with calorie breakdown	Must Have	9	5
US-09	Personalized weekly workout schedule	Must Have	10	6
US-10	Personalized daily meal plan	Must Have	10	7
US-13	Connect wearable device	Should Have	8	8
US-16	Interactive weekly dashboard	Should Have	7	9
US-19	Workout reminders	Should Have	6	10
US-22	Encrypt health data at rest & transit	Must Have	10	11
US-04	Dietary preferences & restrictions	Should Have	7	12
US-07	Log daily water intake	Could Have	5	13
US-08	Food database search	Should Have	7	14
US-11	Auto-adjust plans as metrics change	Should Have	7	15
US-12	Nutritionist review of AI diet plans	Should Have	8	16
US-14	Sync heart rate & sleep data	Could Have	6	17
US-15	Notify on device sync failure	Could Have	4	18
US-17	Trend graphs over time	Could Have	6	19
US-18	Export progress report as PDF	Could Have	5	20
US-20	Meal & hydration reminders	Could Have	5	21
US-21	Customize reminder frequency	Won't Have (this release)	3	22
US-23	Control third-party data access	Should Have	8	23
US-24	Audit logs of data access	Could Have	6	24

Business Value is scored by the Product Owner in consultation with stakeholders; higher scores indicate greater impact on user adherence, safety, or core functionality.

5. Sprint Backlog (Sprint 1)

5.1 Selected Stories for Sprint 1

Based on backlog priority and team capacity, the following high-priority "Must Have" stories are selected to form Sprint 1 (a 2-week sprint): US-01, US-02, US-03, US-05, US-06, US-09, US-10, and US-22. These stories establish the core onboarding, tracking, and recommendation foundation that later epics depend on.

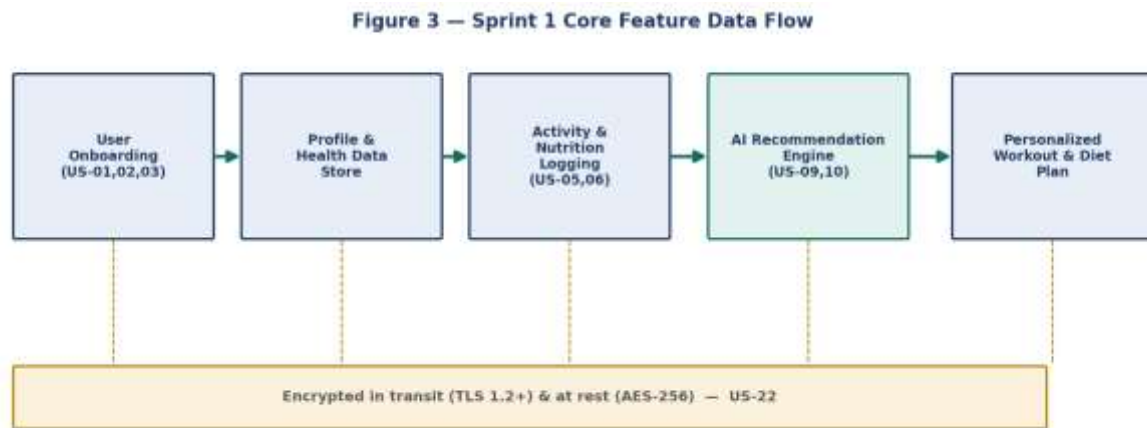


Figure 3: Sprint 1 core feature data flow — from onboarding through to a personalized plan, secured end-to-end.

5.2 Sprint Backlog — Task Breakdown

Task ID	Task Description	Assigned Role	Est. Hours	Linked Story
T-01	Design and implement user registration (email + OAuth)	Backend + Frontend Dev	8 hrs	US-01
T-02	Build profile setup UI capturing age/weight/height	Frontend Dev	6 hrs	US-02
T-03	Implement BMI calculation service and validation rules	Backend Dev	5 hrs	US-02
T-04	Build goal-selection screen (weight loss/gain/maintenance)	Frontend Dev	4 hrs	US-03
T-05	Design activity-log data model and API endpoints	Backend Dev	8 hrs	US-05
T-06	Build workout logging UI (type, duration, intensity)	Frontend Dev	6 hrs	US-05
T-07	Integrate food database API for meal search	Backend Dev	10 hrs	US-06
T-08	Build meal-logging UI with macro breakdown display	Frontend Dev	8 hrs	US-06
T-09	Design initial recommendation engine (rule-based v1)	Data Scientist / ML Eng	14 hrs	US-09
T-10	Build weekly workout-plan generation service	Backend + ML Eng	10 hrs	US-09
T-11	Design diet-plan recommendation model (calorie/macro targeting)	Data Scientist / ML Eng	12 hrs	US-10
T-12	Implement AES-256 encryption at rest and TLS in transit	Security / Backend Dev	8 hrs	US-22
T-13	Write unit & integration tests for onboarding and tracking modules	QA Engineer	10 hrs	Multiple
T-14	Conduct sprint review demo preparation	Scrum Master + Team	3 hrs	All

Total estimated effort: 112 hours across a 2-week sprint for a team of 4 (2 developers, 1 ML engineer, 1 QA), assuming ~6 focused hours/day per member — within team capacity.

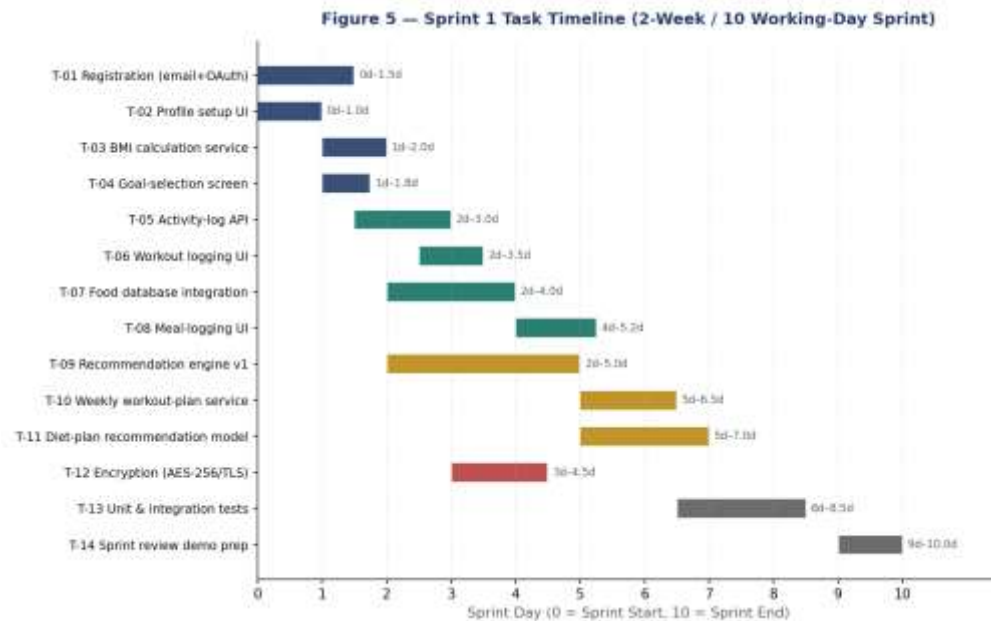


Figure 5: Sprint 1 task timeline across the 10 working days of the sprint.

6. Story Point Estimation

Story points were estimated using Planning Poker with the full development team (2 developers, 1 ML engineer, 1 QA engineer, facilitated by the Scrum Master). Each member privately selected a card from the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21); differences greater than one step were discussed until consensus was reached. Estimates reflect relative complexity, effort, and uncertainty — not absolute hours.

Story ID	User Story (Summary)	Story Points	Justification
US-01	Register using email or social account	3	Standard auth flow with OAuth adds moderate complexity; well-understood pattern, low risk.
US-02	Input age, weight, height, BMI	2	Simple form + calculation logic; minimal dependencies.
US-03	Select fitness goal	1	Single-select UI with no backend complexity.
US-05	Log daily workouts	3	CRUD feature with daily aggregation logic; moderate UI work.
US-06	Log meals with calorie breakdown	5	Requires third-party food database integration and macro aggregation — higher uncertainty.
US-09	Personalized weekly workout schedule	8	Involves new recommendation logic, rule tuning, and edge-case handling for varied user profiles.
US-10	Personalized daily meal plan	8	Similar complexity to US-09 plus dietary-restriction filtering; high uncertainty in first iteration.
US-22	Encrypt health data at rest & transit	5	Security-critical; requires careful implementation and testing though pattern is well documented.

Total Sprint 1 Story Points: 35



Figure 4: Planned sprint burndown — ideal vs. projected actual story points remaining.

The team's historical velocity (from prior comparable sprints) is approximately 32–38 points per 2-week sprint, confirming Sprint 1's scope of 35 points is realistic and within capacity.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, users can register, build a personal health profile, log daily workouts and meals, receive a first personalized workout and diet recommendation, and trust that their health data is securely stored — establishing the core foundation for all future personalization features."

7.2 Expected Sprint Deliverables

- A functional registration and profile-setup flow (email/OAuth login, age/weight/height/BMI capture).
- A working activity and meal logging module with real-time calorie and macronutrient calculation.
- A first-version (v1) rule-based recommendation engine producing a weekly workout plan and daily meal plan.
- End-to-end encryption of all health data at rest (AES-256) and in transit (TLS 1.2+).
- Automated test coverage for onboarding, tracking, and recommendation modules (target $\geq 80\%$).
- A sprint review demo showcasing the onboarding-to-first-recommendation user journey.

7.3 Definition of Done

- Code peer-reviewed and merged to the main branch.
- Unit and integration tests written and passing in CI.
- Acceptance criteria for each story verified by QA against the Given-When-Then scenarios in Section 3.
- No critical or high-severity security defects open.
- Feature demoed and accepted by the Product Owner in the Sprint Review.

7.4 Next Sprint Preview

Sprint 2 will prioritize wearable device integration (US-13, US-14, US-15), dashboard visualization (US-16, US-17), and reminder notifications (US-19, US-20) — building on the tracking and recommendation foundation delivered in Sprint 1.